Based on the configuration:

```
Recurring fee: $100/month
Due day: 4th of month
Grace period: 1 day
Partial payments: Allowed
Late fees: Enabled

Initial fee: 10%
Daily fee: Yes
Max cap: $30
```

The late fee workflow should work as follows.

Invoice Timeline Example

Invoice Generated

```
Invoice Amount: $100
Due Date: July 4, 2026
```

Status:

```
Due
Outstanding Balance: $100
Late Fee: $0
```

Grace Period

Since:

```
Grace Period = 1 day
```

The owner has until:

```
None
July 5, 2026 11:59 PM
America/New_York
```

to make payment without any penalty.

During:

```
None
July 4
July 5
```

No late fee should be charged.

Late Fee Trigger

Beginning:

```
None
July 6, 2026
```

The invoice becomes eligible for late fees.

Initial Late Fee

Initial fee:

```
None
10%
```

Invoice amount:

```
None
$100
```

Late fee:

```
None
$10
```

The system should automatically create a transaction/activity entry:

```
None
Late Fee Applied
Amount: $10
Date: July 6, 2026
Reason: Initial Late Fee (10%)
```

Invoice becomes:

```
None
Original Invoice: $100
Late Fee: $10
Total Due: $110
```

Daily Late Fee

Because:

```
None
Daily Fee = Yes
```

The system should continue adding late fees daily until:

- Invoice is fully paid
- Maximum cap is reached

Example

July 7

```
None
+$1 Daily Late Fee
```

Total:

```
None
$111
```

Transaction:

```
None
Daily Late Fee Applied
Amount: $1
```

July 8

```
None
+$1 Daily Late Fee
```

Total:

```
None
$112
```

Continue until:

```
None
Total Late Fees = $30
```

(Max Cap)

Maximum Cap

Configured:

```
None
Max Cap = $30
```

Therefore:

```
None
Initial Fee = $10
```

```
Daily Fees = Up to $20 additional
```

Once late fees reach:

```
None
$30
```

The system must stop adding additional late fees.

Example:

```
None
Invoice Amount: $100
Late Fees: $30
Total Due: $130
```

No more late-fee transactions should be created.

Partial Payment Scenario

Because:

```
None
Partial Payments = Allowed
```

Example:

Owner Pays $50

Outstanding:

```
None
Invoice Amount: $100
Paid: $50
Remaining: $50
```

Status:

```
None
Partially Paid
```

Late fees should continue to be calculated on the unpaid balance according to your business rules.

Recommended approach:

```
None
Calculate late fees on remaining unpaid balance
```

Instead of the original invoice amount.

Example:

```
None
Remaining Balance = $50
Initial Late Fee = $5 (10%)
```

This is generally fairer and more accurate.

Invoice Activity Timeline

Every late fee application should create an auditable transaction.

Example:

```
None
July 6
Late Fee Applied (+$10)

July 7
Daily Late Fee Applied (+$1)

July 8
Daily Late Fee Applied (+$1)

July 9
Owner Payment Received ($50)

July 10
```

```
Daily Late Fee Applied (+$1)
```

Invoice Detail Page

A dedicated section should show:

```
None
Late Fee Summary

Original Invoice: $100
Initial Late Fee: $10
Daily Late Fees: $5
Total Late Fees: $15

Current Amount Due: $115
```

Downloaded Invoice / Statement

The PDF should show late fees as separate line items:

```
None
Description                    Amount

Monthly HOA Fee               $100.00
Initial Late Fee (10%)         $10.00
Daily Late Fee                  $1.00
Daily Late Fee                  $1.00

Total Due                     $112.00
```

Expected Outcome

If the backend and UI are configured correctly:

- Late fees should automatically trigger after the grace period.
- Every late fee application should create an invoice transaction/activity record.
- Late fees should appear in:

    - Invoice Detail Page
    - Invoice Timeline
    - Downloaded Invoice PDF
    - Payment Calculations
    - Owner Invoice View
- Late fees should stop automatically once the configured **$30 maximum cap** is reached.